

Linear Algebra Homework

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Question 1

$$\vec{u} = \langle 1, 4, -7 \rangle, \quad \vec{v} = \langle 2, -1, 4 \rangle, \quad \vec{w} = \langle 1, 5, -1 \rangle, \quad \vec{b} = \langle 0, -9, 18 \rangle$$

(a) **Computing the area of the parallelogram determined by the vectors $\vec{u}$ and $\vec{v}$**

The area determined by two vectors is given by the magnitude of their cross product

$$\begin{aligned} \|\vec{u} \times \vec{v}\| &= \|\langle u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1 \rangle\| \\ \|\vec{u} \times \vec{v}\| &= \|\langle 16 - 7, -14 - 4, -1 - 8 \rangle\| \\ \|\vec{u} \times \vec{v}\| &= \|\langle 9, -18, -9 \rangle\| \\ \|\vec{u} \times \vec{v}\| &= \sqrt{(9)^2 + (-18)^2 + (-9)^2} \\ \|\vec{u} \times \vec{v}\| &= \sqrt{486} \end{aligned}$$

(b) **Computing the volume of the parallelepiped determined by the vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$**

The volume of a parallelepiped determined by three vectors is the absolute value of the cross product of two vectors dotted with the third vector

$$\begin{aligned} |(\vec{u} \times \vec{v}) \cdot \vec{w}| &= |\langle 9, -18, -9 \rangle \cdot \langle 1, 5, -1 \rangle| \\ |(\vec{u} \times \vec{v}) \cdot \vec{w}| &= |9 - 90 + 9| \\ |(\vec{u} \times \vec{v}) \cdot \vec{w}| &= 72 \end{aligned}$$

(c) **Finding $|\vec{b} \cdot (\vec{u} \times \vec{v})|$**

$$\begin{aligned} |\vec{b} \cdot (\vec{u} \times \vec{v})| &= |\langle 0, -9, 18 \rangle \cdot \langle 9, -18, -9 \rangle| \\ &= 0 + 162 - 162 \\ &= 0 \end{aligned}$$

Thus, we can conclude that the vectors $\vec{u}$, $\vec{v}$ and $\vec{b}$ are coplanar

Question 2

Finding the vector equation of a line given a point and a parallel line

We are given a parallel line so finding a vector between any two points on the line is the direction vector for the other line

Getting the vector between $t = 0$ and $t = 1$ for the parallel line

$$\begin{aligned} \vec{v} &= \langle -1 - 2, 3 + 1, 7 - 8 \rangle \\ &= \langle -3, 4, -1 \rangle \end{aligned}$$

We now have both a point on the line and a direction vector and thus are able to find the equation of the line

$$\begin{aligned}\vec{r} &= \langle -1, 2, 4 \rangle + t \langle -3, 4, -1 \rangle \\ &= \langle -3t - 1, 4t + 2, 4 - t \rangle\end{aligned}$$

Question 3

Finding the equation of a plane given a point and a perpendicular line

A plane is described by a point in the plane and a normal vector

We are given a line perpendicular to the plane, and so a direction vector of the line is the normal vector of the plane

Take the direction vector between $t = 0$ and $t = 1$ as our normal vector

$$\begin{aligned}\vec{n} &= \langle -3, 2 - 1, 3 - 7 \rangle \\ &= \langle -3, 1, -4 \rangle\end{aligned}$$

Let $P(x, y, z)$ be any other point on the plane and let $P_0(2, 0, 1)$ be the known point. Then,

$$\overrightarrow{P_0P} \cdot \vec{n} = 0$$

$$\begin{aligned}\langle x - 2, y, z - 1 \rangle \cdot \langle -3, 1, -4 \rangle &= 0 \\ 6 - 3x + y + 4 - 4z &= 0 \\ 3x - y + 4z &= 10\end{aligned}$$

Thus, $3x - y + 4z = 10$ is the equation of the plane

Question 4

Finding the equation of a plane which passes through three given points

$$P_1 = (1, 1, 0), \quad P_2 = (2, 0, 1), \quad P_3 = (0, -1, 3)$$

To describe the plane, we need a normal vector, which we can obtain by getting the cross product of two vectors in the plane

$$\overrightarrow{P_1P_2} = \langle 1, -1, 1 \rangle, \quad \overrightarrow{P_2P_3} = \langle -2, -1, 2 \rangle$$

$$\begin{aligned}\vec{n} &= \overrightarrow{P_1P_2} \times \overrightarrow{P_2P_3} \\ &= \langle -2 + 1, -2 - 2, -1 - 2 \rangle \\ &= \langle -1, -4, -3 \rangle\end{aligned}$$

Let $P(x, y, z)$ be any other point in the plane

$$\begin{aligned}\overrightarrow{PP_1} \cdot \vec{n} &= 0 \\ \langle x - 1, y - 1, z \rangle \cdot \langle -1, -4, -3 \rangle &= 0 \\ 1 - x, 4 - 4y, -3z &= 0 \\ x + 4y + 3z &= 5\end{aligned}$$

Thus, $x + 4y + 3z = 5$ is the equation of the plane

Question 5

Finding the intersection of a line and a plane

Let the vector between the two points ($\vec{v}$) be the direction vector

$$\vec{v} = \langle 3, -2, 1 \rangle$$

Then the equation of the line is

$$\begin{aligned}\vec{r} &= \langle 1, 0, 1 \rangle + t\langle 3, -2, 1 \rangle \\ &= \langle 1 + 3t, -2t, 1 + t \rangle\end{aligned}$$

The parametric equations are

$$x = 1 + 3t$$

$$y = -2t$$

$$z = 1 + t$$

Substituting these into the equation of the plane, to find the point of intersection

$$x + y + z = 6$$

$$1 + 3t - 2t + 1 + t = 6$$

$$t = 2$$

The point at $t = 2$ is $(7, -4, 3)$, which is the point of intersection

Question 6

Do three given points form a right triangle?

$$P_1 = (-1, 7, 9), \quad P_2 = (1, 5, 14), \quad P_3 = (-9, 11, 13)$$

Any three points in $\mathbb{R}^3$ are coplanar, and such a triangle is formed between them

So the only thing to check is whether or not there is a right angle

Check this by checking if any two of the vectors between the three points are perpendicular to each other

$$\overrightarrow{P_1P_2} = \langle 2, -2, 5 \rangle, \quad \overrightarrow{P_2P_3} = \langle -10, 6, -1 \rangle, \quad \overrightarrow{P_1P_3} = \langle -8, 4, 4 \rangle$$

$$\begin{aligned}\overrightarrow{P_1P_2} \cdot \overrightarrow{P_2P_3} &= -20 - 12 - 5 \\ &= -37 \\ &\neq 0\end{aligned}$$

$$\begin{aligned}\overrightarrow{P_1P_2} \cdot \overrightarrow{P_1P_3} &= -16 - 8 + 20 \\ &= -4 \\ &\neq 0\end{aligned}$$

$$\begin{aligned}\overrightarrow{P_1P_3} \cdot \overrightarrow{P_2P_3} &= -80 + 24 - 4 \\ &= -60 \\ &\neq 0\end{aligned}$$

Since none of the three vectors are perpendicular to one another, we conclude that a right triangle is not formed

Question 7

Find three points that do form a right triangle in $\mathbb{R}^3$

Let,

$$P_1 = (-1, 7, 9), \quad P_2 = (1, 5, 14), \quad P_3 = (x, y, z)$$

Then,

$$\overrightarrow{P_1P_2} = \langle 2, -2, 5 \rangle, \quad \overrightarrow{P_2P_3} = \langle x-1, y-5, z-14 \rangle, \quad \overrightarrow{P_1P_3} = \langle x+1, y-7, z-9 \rangle$$

For there to be a right angle any two vectors should be perpendicular

$$\overrightarrow{P_1P_2} \cdot \overrightarrow{P_2P_3} = 0$$

$$2x - 2 - 2y + 10 + 5z - 70 = 0$$

$$2x - 2y + 5z = 62$$

For the three points to form a right triangle, P_3 needs to be in the plane $2x - 2y + 5z = 62$

One such point is

$$P_3 = (31, 0, 0)$$

Thus the points $(-1, 7, 9)$, $(1, 5, 14)$ and $(31, 0, 0)$ form a right triangle